

The S3 Closure Pool Against Its Own Definition

Recomputing every interaction score and every p-value from the singles, doubles and triples the 025
build joined

Michael Volk

September 18, 2026

Abstract

The S3 closure pool holds, for every one of its 376,732 measured triples, every term of the equation that defines the triple's interaction score: the three singles, the three doubles and the triple itself, 1,121,645 records in all. Inside any one source screen that equation is exact: recomputing the digenic score from the fitness columns of the same row reproduces the reported score at $r = 0.999$ in Costanzo 2016 and $r = 1.000$ in Kuzmin 2018. Inside the build it is not. The build's own fitness reproduces its own digenic scores at $r = 0.445$ and its own trigenic scores at $r = 0.230$, and the loss is traceable to the join rather than to the formula. Four mechanisms account for it. SGD essentiality enters as a fitness of 0 and is averaged with the measured fitness of the same gene's allele, which pulls 907 singles down by a median of 0.16 and shifts every score built on them by 0.12; taking the 0 back out halves the digenic error on the 139,461 doubles those singles touch. Screens are averaged across their own controls, so a single that Costanzo measured against 350 control screens is combined with the same gene's Kuzmin estimate and then used in an identity that was only ever true within one screen. Duplicate screens of the same genotype, 94 percent of the closure doubles, are merged into one record whose p-value is a t -test over the two or three duplicate scores with one or two degrees of freedom, which discards the source p-values entirely: the merged p correlates with the median source p at $\rho = 0.05$, and 1,754 doubles significant in every source are not significant in the build. And 99.6 percent of the Kuzmin 2020 digenic rows carry no query fitness, which the source scoring replaced by 1.0 before computing the score, a substitution that is recorded nowhere in the released table but is recovered here exactly. Where the build kept a single screen, its p-value is the source's and is reproducible from the double's own colony standard deviation at $\rho = 0.984$; the trigenic p-values are not reproducible from anything the build holds, because their null variance comes from control screens that were never released. The consequence for training on S3 is stated at the end: the fitness the model is asked to learn and the interaction it is asked to learn are not consistent with each other in this build, so the identity cannot be the mechanism by which fitness supervision helps, and every p-value in a merged record should be treated as unlabeled.

Section status:  ready  author-reviewed, keep stable  not done, agents may edit freely

Contents

1	The question 	2
2	How the pool was built 	2
3	Strength: the score against its own definition 	4
4	Confidence: the p-value against what the build holds 	6
5	The hazards of the join, counted 	7
6	What this means for training on S3 	8
7	Limits 	9

1 The question x

A genetic interaction score is not a measurement. It is a difference between a measured fitness and a fitness expected from lower-order measurements, and the expectation is a product. For two deletions a and b with single-mutant fitness f_a and f_b and double-mutant fitness f_{ab} ,

$$\varepsilon_{ab} = f_{ab} - f_a f_b, \quad (1)$$

and for three, with the three doubles already scored,

$$\tau_{abc} = f_{abc} - f_a f_b f_c - \varepsilon_{ab} f_c - \varepsilon_{ac} f_b - \varepsilon_{bc} f_a. \quad (2)$$

Both equations are stated verbatim in the supplementary text of Kuzmin 2018 and Kuzmin 2020 (mirror keys `kuzminSystematicAnalysisComplex2018`, `kuzminExploringWholegenomeDuplicate2020`, `si/si1.md` lines 84 to 91 and 64 to 67), and Eq. 1 in Costanzo 2016 and Baryshnikova 2010. Substituting Eq. 1 into Eq. 2 gives the form used here, $\tau_{abc} = f_{abc} - f_a f_b f_c - f_{ab} f_c - f_{ac} f_b - f_{bc} f_a + 2f_a f_b f_c$.

The S3 closure pool of the 025 build was assembled so that a model trained on it sees every term of Eq. 2 for every triple it is asked to score: all 5,694 single-deletion records in the build, the 739,219 doubles whose gene pair lies inside some measured triple, and all 376,732 triples. The training arm (`cgt_s3_r_kl_fit_031`, three seeds on IGB `mmli`) supervises fitness and interaction jointly, with the 010 random split pinned over the triples. That design carries an assumption worth testing before the runs finish: that the fitness values in the pool and the interaction values in the pool are consistent with each other under the equations that define the latter.

Two questions follow, and the document answers both against the source tables the build was assembled from.

Strength. Does the fitness stored in the build reproduce the interaction stored in the build? Eq. 1 on every closure double and Eq. 2 on every triple, the recomputed score against the stored one, and the same recomputation on the raw source rows as a control for the formula.

Confidence. Does anything stored in the build reproduce the p-value stored in the build? Each interaction record carries a p-value from its source screen, or, when the build merged several screens of the same genotype, a p-value the build computed itself. The recomputation uses the stored standard deviations under the normal model the sources describe, and for merged records compares the stored p against the p-values the sources reported for the same genotype.

Three earlier readings frame the answer. The recapitulation of 2026-09-04 (`recapitulate_tmi_from_fitness.py`) put the trigenic identity on the whole build at $r = 0.23$ and concluded that noise accumulation, not a sign or scale error, was the cause. The 025 loaders' provenance work established that the released combined-mutant standard deviation is a colony sample SD over four colonies and that a two-sided normal test on it ranks the reported p-values at $\rho = 0.985$. And the additive-baselines report established that a per-gene additive ridge reaches $r = 0.40$ on these labels without any fitness at all. What is new here is the decomposition: which step of the join costs what, measured on the pool a model is about to be trained on.

2 How the pool was built x

Figure 1 traces a record from its source screen to the trainer. Each step is a committed piece of code, so each hazard has a line number rather than a suspicion.

The sources. Costanzo 2016 contributes single-mutant fitness at 26 and 30 degrees, double-mutant fitness with a colony standard deviation, and a digenic score with a p-value, one row per query-array pair per screen, 12.2 million double rows over four released files. Its single-mutant fitness is not screen-specific: the supplementary text (`costanzoGlobalGeneticInteraction2016/si/si1.md` line 94) states that array strains were “based on an average of 350 replicate control screens” and query strains on 17, so f_a and f_b in Eq. 1 are genome-wide standards. Kuzmin 2018 and Kuzmin 2020 contribute single, double and triple fitness, digenic and trigenic scores and their p-values, one row per query-array pair, 501 thousand and 1.4 million rows. Their array single-mutant fitness is taken from Costanzo 2016 (`kuzminSystematicAnalysisComplex2018/si/si1.md` line 135: “single mutant fitness estimates available from a previous study”), and the two digenic terms of Eq. 2 come from the single-mutant control query screened “in the same batch” as the double-mutant query (line 75). SGD essentiality contributes a label with no fitness measurement, and SynthLethDB a synthetic-lethal pair with none.

The build. `CompositeFitnessConverter` turns an essentiality or synthetic-lethality entry into a fitness of 0. `MeanExperimentDeduplicator` (`torchcell/data/mean_experiment_deduplicate.py`) then merges every entry that shares an experiment type and a sorted gene set: fitness becomes the mean and its standard deviation the root mean square (lines 308 to 316); an interaction score becomes the mean, and its p-value becomes a two-sided t -test of the

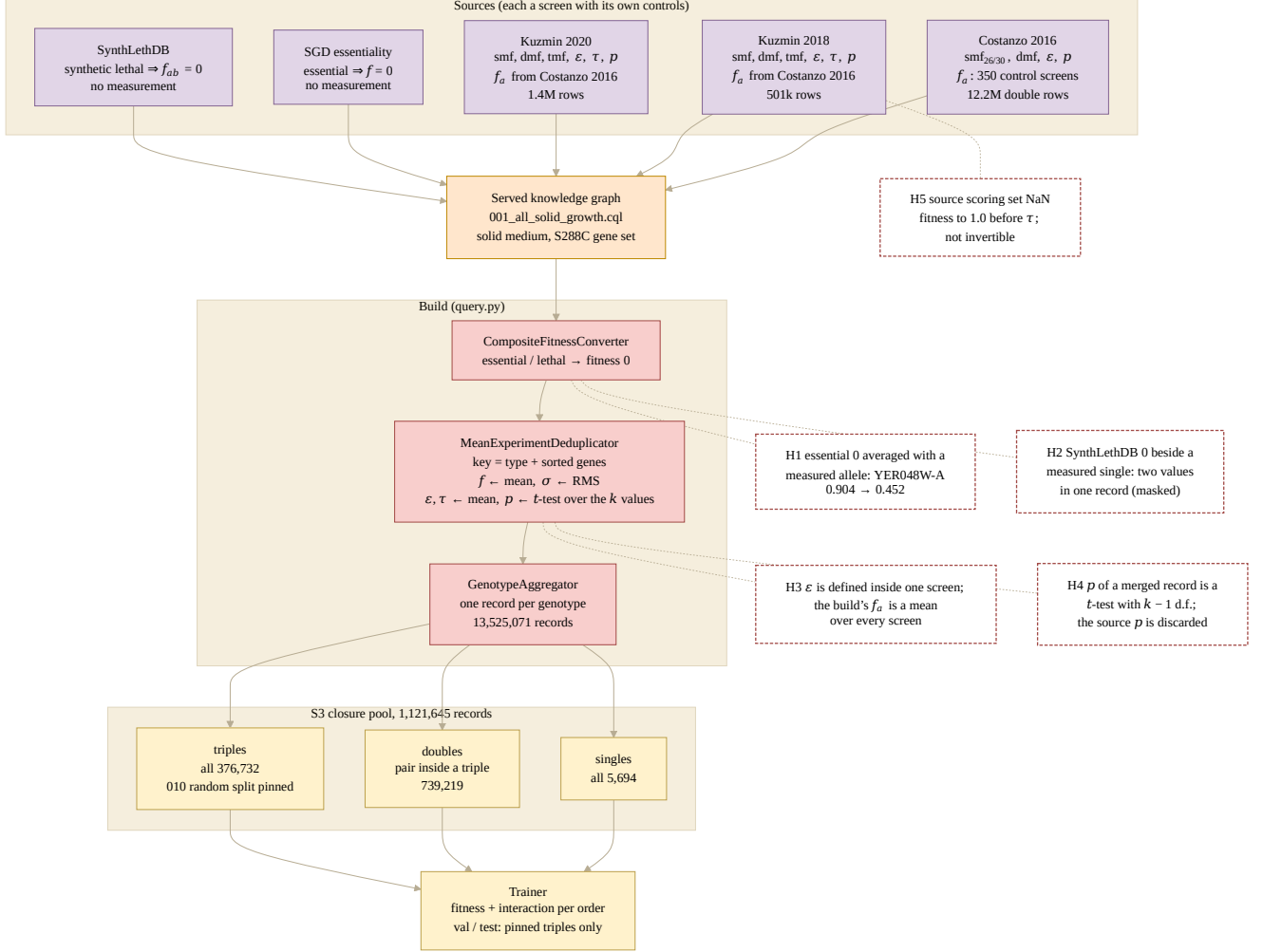


Figure 1: From five sources to one training pool. Purple: the sources, each a screen with its own controls and its own single-mutant fitness standard. Orange: the Cypher query over the served knowledge graph (`experiments/025-solid-growth/queries/001_all_solid_growth.cql`), filtered to solid medium and the S288C gene set. Red: the build (`experiments/025-solid-growth/scripts/query.py`): essentiality and synthetic lethality become a fitness of 0, duplicates of one experiment type and one gene set are merged into a mean, and records are grouped per genotype. Yellow: the S3 pool and the trainer. Dashed boxes H1 to H5 are the hazards quantified in Sections 3 to 5; each attaches to the step that introduces it.

merged scores against zero with $k - 1$ degrees of freedom (lines 480 to 494), which does not read the source p-values at all. **GenotypeAggregator** groups what remains into one record per genotype, 13,525,071 of them. The trainer reads one fitness value and one interaction value per record; a record that still carries two fitness values, which happens when a SynthLethDB entry lists the same gene twice and so escapes the merge key, is masked (three records, Section 5).

The pool. Table 1 lists what the join produced, by perturbation order and by the set of sources behind each record’s fitness. Of the 5,694 singles, 1,140 carry an SGD essentiality entry, and 233 of those carry nothing else. Of the 739,219 closure doubles, 697,825 (94.4 percent) are merged from two or more screens, most often the two orientations and two temperatures at which Costanzo screened the same pair. Of the 376,732 triples, 12,914 are merged, mostly the same triple reached from two different double-mutant queries.

Table 1: Records of the S3 pool by perturbation order and by the sources joined into their fitness value. Generated by `s3_closure_recompute.py` (`results/s3_closure_composition.csv`). “GeneEssentialitySgd” means an essentiality entry converted to a fitness of 0 entered the mean.

order	fitness sources joined	records
1	SmfCostanzo2016	3,193
	SmfCostanzo2016 + SmfKuzmin2018	1,111
	GeneEssentialitySgd + SmfCostanzo2016	673
	GeneEssentialitySgd + SmfCostanzo2016 + SmfKuzmin2018	234
	GeneEssentialitySgd	233
	SmfCostanzo2016 + SmfKuzmin2020	185
	SmfCostanzo2016 + SmfKuzmin2018 + SmfKuzmin2020	41
	SmfKuzmin2018	12
	SmfKuzmin2020	9
	SmfCostanzo2016 + SynthLethalityYeastSynthLethDb	2
2	SmfCostanzo2016 + SmfKuzmin2018 + SynthLethalityYeastSynthLethDb	1
	DmfCostanzo2016 + DmfKuzmin2020	514,325
	DmfCostanzo2016 + DmfKuzmin2018	166,377
	DmfKuzmin2020	37,148
	DmfCostanzo2016 + DmfKuzmin2018 + DmfKuzmin2020	14,791
	DmfKuzmin2018	5,358
	DmfCostanzo2016 + DmfKuzmin2020 + SynthLethalityYeastSynthLethDb	351
	DmfKuzmin2018 + DmfKuzmin2020	321
	DmfCostanzo2016 + DmfKuzmin2018 + SynthLethalityYeastSynthLethDb	277
	DmfCostanzo2016	208
	DmfCostanzo2016 + DmfKuzmin2018 + DmfKuzmin2020 + SynthLethalityYeastSynthLethDb	31
	DmfCostanzo2016 + SynthLethalityYeastSynthLethDb	17
	DmfKuzmin2020 + SynthLethalityYeastSynthLethDb	8
	DmfKuzmin2018 + SynthLethalityYeastSynthLethDb	5
	SynthLethalityYeastSynthLethDb	2
3	TmfKuzmin2020	285,682
	TmfKuzmin2018	90,567
	TmfKuzmin2018 + TmfKuzmin2020	483

3 Strength: the score against its own definition ✗

The formula is exact inside a screen. Figure 2a and the first three rows of Table 2 are the control. On the raw Costanzo 2016 rows the identity reproduces the reported score at $r = 0.999$, slope 1.00, root-mean-square error 0.002; on the raw Kuzmin 2018 rows at $r = 1.000$ with zero residual; on the raw Kuzmin 2020 rows at $r = 0.996$. The Kuzmin 2020 figure needs one substitution to reach that: 99.6 percent of its digenic rows carry no query single-mutant fitness, and the identity reproduces the reported score only when that missing value is set to 1.0, which is what the supplementary text says the scoring did (“all NaN fitness estimates were assigned a value of 1.0”, `si/si1.md` line 41). Hazard H5 is therefore not a hypothesis: the reported Kuzmin 2020 digenic scores were computed against a query fitness of 1.0, and the build inherits scores that no stored fitness can reproduce for those pairs. The small vertical streak at a reported score of 0 in Figure 2a is the 2.1 percent of Costanzo rows with a missing query fitness, where the same rule applies.

Digenic, in the build. The same identity on the build’s own values reaches $r = 0.445$, $\rho = 0.493$, slope 0.99 and a root-mean-square error of 0.072 over 728,856 doubles (Figure 2b). The slope says the identity holds in expectation; the error says it fails in value, by an amount larger than the stored score’s own spread. The strata separate the causes. Doubles whose two singles carry no essentiality entry reach $r = 0.598$ with error 0.042; doubles touching an essentiality-tainted single reach $r = 0.452$ with error 0.140 and an intercept of 0.116, which is the signature of a product $f_a f_b$ that is too small because one factor was averaged with a 0. Restoring the measured mean of the tainted single moves that

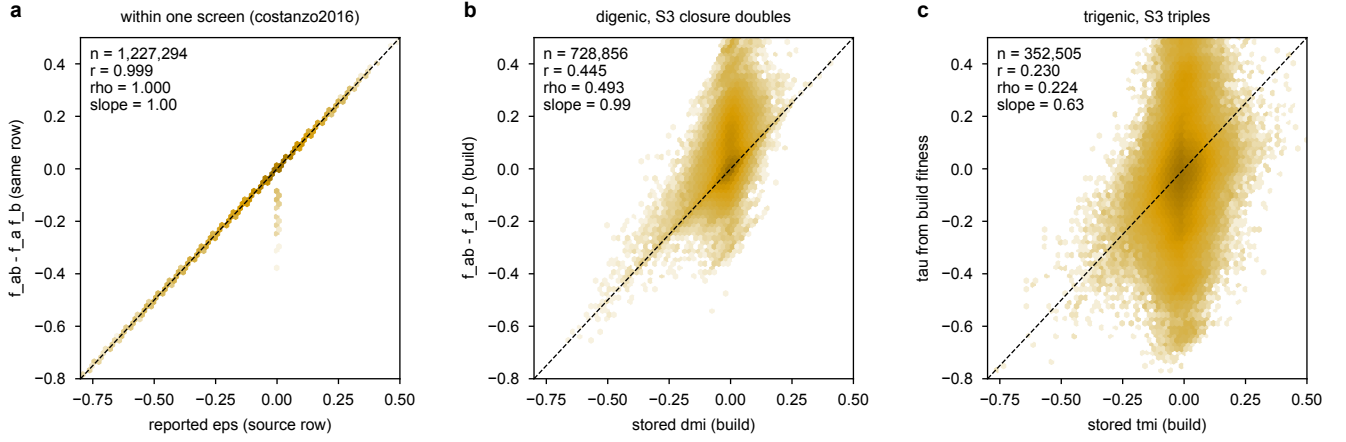


Figure 2: Recomputed against stored interaction scores. (a) Within one source screen: each Costanzo 2016 row’s $f_{ab} - f_a f_b$ against the score reported on the same row, over the 1,227,294 rows whose gene pair is a closure pair. (b) In the build: Eq. 1 from the build’s single and double fitness against the stored digenic score, all 728,856 closure doubles with both. (c) In the build: Eq. 2 from the build’s fitness against the stored trigenic score, the 352,505 triples with full closure. Hexagon shade is log count; the dashed line is identity.

Table 2: Recomputed against stored score, by stratum. Within-screen rows recompute from the columns of one raw source row; every other row recomputes from the build’s stored fitness. “Measured mean restored” takes the essentiality 0 back out of each tainted single by $k/(k-1)$ (Section 5). The best r is bold. Generated by `s3_closure_recompute.py`.

stratum	n	r	ρ	slope	rmse
within one screen, Costanzo 2016 (raw row)	1,227,294	0.999	1.000	1.00	0.002
within one screen, Kuzmin 2018 (raw row)	197,876	1.000	1.000	1.00	0.000
within one screen, Kuzmin 2020 (raw row)	601,330	0.996	0.997	1.00	0.004
digenic, all closure doubles	728,856	0.445	0.493	0.99	0.072
digenic, one screen only	36,499	0.339	0.314	0.95	0.120
digenic, merged screens	692,357	0.459	0.509	0.99	0.068
digenic, no essential single	589,395	0.598	0.633	0.98	0.042
digenic, an essential single (stored mean)	139,461	0.452	0.382	0.97	0.140
digenic, an essential single (measured mean restored)	139,461	0.491	0.441	0.96	0.071
trigenic, all triples	352,505	0.230	0.224	0.63	0.171
trigenic, one screen only	339,616	0.239	0.233	0.66	0.169
trigenic, merged screens	12,889	0.114	0.153	0.30	0.223
trigenic, no essential single	278,060	0.205	0.211	0.58	0.169
trigenic, an essential single (stored mean)	74,445	0.255	0.188	0.60	0.180
trigenic, an essential single (measured mean restored)	74,445	0.250	0.186	0.60	0.167
trigenic, from stored dmi	350,318	0.213	0.174	0.48	0.144

stratum to $r = 0.491$, intercept 0.014, error 0.071: half the error of that stratum was the essentiality 0. Doubles kept from one screen reach only $r = 0.339$, against $r = 0.459$ for merged doubles, because a merged double's fitness and score are both means over screens and average some of the same noise away, while a single-screen double's fitness is compared against singles averaged over other screens.

Trigenic, in the build. Eq. 2 on the build's fitness reaches $r = 0.230$, slope 0.63 and error 0.171 over the 352,505 triples whose three singles and three doubles are all present (Figure 2c). The error is 2.7 times the standard deviation of the stored trigenic score, so the reconstruction carries more propagated measurement error than signal, which is the attenuation the 2026-09-04 recapitulation computed. The essentiality strata behave as in the digenic case (intercept -0.085 with the stored singles, 0.022 with the measured mean restored) without moving r , because five terms of noise remain. Triples merged from two queries reach $r = 0.114$, and using the stored digenic scores in place of the recomputed products gives $r = 0.213$, no rescue. At the intermediate threshold of $|\tau| > 0.08$ the build's fitness calls 169,476 triples where the stored scores call 46,889, and the two agree on 29,327, a recall of 0.63 and a precision of 0.17.

4 Confidence: the p-value against what the build holds $\times$

What a p-value in the build is. Two kinds of p-value sit under the same field. A record kept from one screen carries the source's p-value unchanged: for every one of the 41,392 single-screen closure doubles and 363,818 single-screen triples, the stored p equals the source row's p to the last digit. A record merged from k screens carries a p-value the build computed, a two-sided t -test of the k merged scores against zero with $k - 1$ degrees of freedom. Every source p-value that went into that record is gone.

The sources' own p-values are not recomputable from anything they released. All four papers describe the null as a pooled variance: Baryshnikova 2010 combines an array-strain term from wild-type control screens with a query-strain term pooled across the screen (si/si1.md lines 262 to 288), and Kuzmin estimates the array term from 91 wild-type control screens (line 63). The released tables carry only the combined mutant's own colony standard deviation. The one formula written out explicitly, for Kuzmin 2020's query-query interactions (si/si1.md line 52), is a one-sided normal test of the score against the propagated variance of the expected product. What can be tested is whether that normal model, on the standard deviations the build stores, ranks the stored p-values.

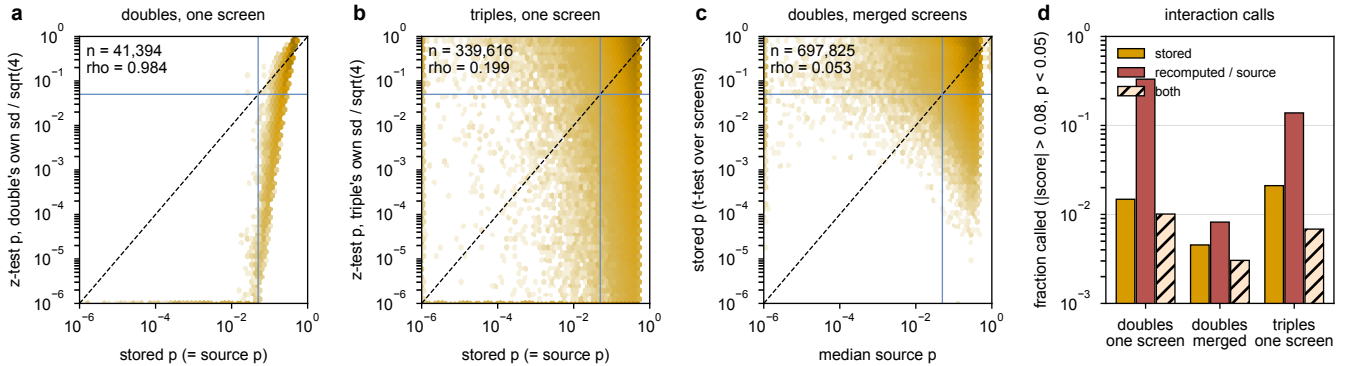


Figure 3: Stored against recomputed p-values, log-log, clipped at 10^{-6} ; blue lines at $p = 0.05$. (a) Single-screen closure doubles: the source p against a two-sided normal test of the stored score with null standard deviation $\sigma_{ab}/\sqrt{4}$, the double's own colony SD over its four colonies. (b) Single-screen triples, the same test with the triple's own SD. (c) Merged closure doubles: the stored t -test p against the median of the p-values the sources reported for the same pair. (d) Fraction of records called an interaction at the intermediate threshold ($|\text{score}| > 0.08$, $p < 0.05$) by the stored fields, by the recomputed score and p (a, b) or the source median (c), and by both; log scale.

A single-screen digenic p is the double's own colony SD. On the 41,394 single-screen closure doubles the normal test on $\sigma_{ab}/\sqrt{4}$ ranks the source p at $\rho = 0.984$ (Figure 3a), the same ordering the loader provenance found on the Kuzmin raw table. Its calls contain 606 of the source's 613 at the intermediate threshold. The test is not the source's test: it calls 1,505 doubles, 2.5 times the source, because the source's pooled null is wider than one double's four colonies. Adding the singles' stored standard deviations to the null lowers the ranking to $\rho = 0.813$, so the stored single-mutant SD is not the quantity the source propagated. Applying the same test to the recomputed score instead of the stored one calls 13,705 doubles, 22 times the source, with 418 in common: the recomputed score carries the join's error but the test does not know it.

A trigenic p is reproducible from nothing in the build. On the 339,616 single-screen triples the same test on the triple's own SD ranks the source p at $\rho = 0.199$ (Figure 3b); propagating every term of Eq. 2 through the stored

Table 3: Stored against recomputed p-values, by stratum. ρ is Spearman’s correlation between the two p-values; the last two columns are the fraction of the stratum called an interaction at $|\text{score}| > 0.08$ and $p < 0.05$ under the stored fields and under the comparison. Generated by `s3_closure_recompute.py`.

comparison	n	ρ of p	called stored	called other
doubles, one screen: source p vs z from the double’s sd	41,394	0.984	0.015	0.036
doubles, one screen: source p vs z with the singles’ sd propagated	41,394	0.813	0.015	0.011
doubles, one screen: source call vs call on recomputed ε	41,394	0.984	0.015	0.331
doubles, merged: stored (t -test) vs z from the double’s sd	697,825	0.686	0.005	0.017
doubles, merged: stored (t -test) vs median source p	697,825	0.053	0.005	0.008
triples, one screen: source p vs z from the triple’s sd	339,616	0.199	0.021	0.110
triples, one screen: source p vs z with every term propagated	339,616	0.097	0.021	0.138
triples, merged: stored (t -test) vs source p	12,914	—	0.067	0.057

SDs gives $\rho = 0.097$. The trigenic null variance is dominated by the two digenic control-screen terms, whose variance was estimated from the same-batch control screens and not released, so the triple’s colony SD carries almost none of it. The test on the stored τ recalls 0.91 of the source calls but makes 5.2 times as many; on the recomputed τ it agrees with the source on 2,323 of its 46,957 calls.

A merged p is a different statistic. On the 697,825 merged closure doubles the stored t -test p correlates with the median source p at $\rho = 0.053$ (Figure 3c), and with the normal test on the merged double’s SD at $\rho = 0.686$. The t -test measures agreement among the k duplicate scores, not the score against measurement noise, and with one or two degrees of freedom it is both too lenient and too strict: 5.4 percent of merged doubles are stored as $p < 0.05$ against 2.0 percent by the source median, while 1,754 doubles significant in every one of their sources are stored as not significant. At the intermediate threshold the stored fields call 3,169 merged doubles and the source medians 5,719, with 2,125 in common. Among the 12,914 merged triples, 505 are significant in every source and 406 of those are stored as not significant.

5 The hazards of the join, counted x

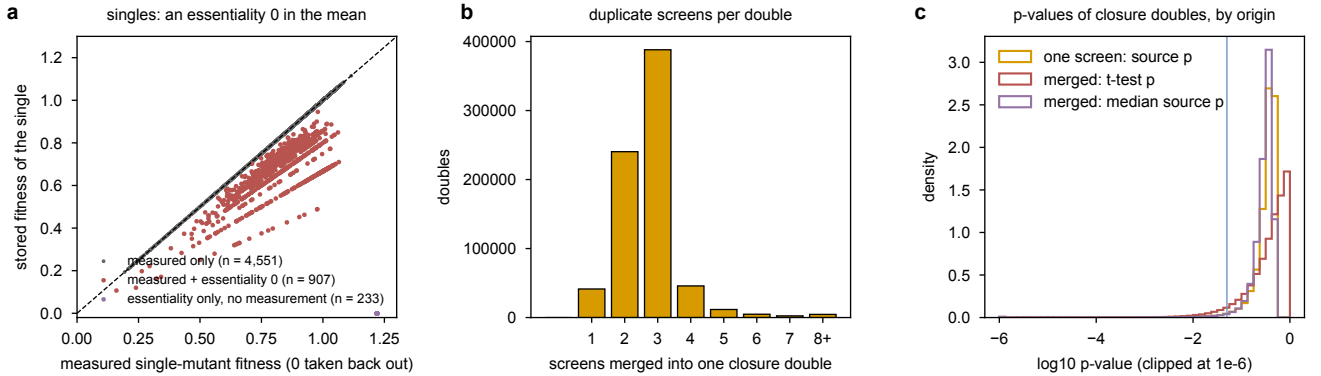


Figure 4: What the join does to the values. (a) Every single in the pool: stored fitness against the measured fitness with the essentiality 0 taken back out ($k/(k-1)$ times the stored mean for a single merged from k entries of which one is the 0). Gray, no essentiality entry, on the identity; red, 907 singles whose measured allele fitness was averaged with a 0; purple, 233 singles with an essentiality entry and no measurement, stored at 0 and placed at the right margin. (b) Number of source screens merged into each closure double. (c) Distribution of $\log_{10} p$ over closure doubles by how the p-value arose: the source’s own p for single-screen records, the build’s t -test for merged records, and the median source p for the same merged records.

H1: an essentiality label averaged into a measured fitness. The SGD essentiality entry is a statement about the gene, not a measurement of a strain. Costanzo and Kuzmin measured essential genes through temperature-sensitive or decreased-abundance alleles that grow, so the same gene name carries a measured allele fitness and a converted 0, and the deduplicator’s key, experiment type plus gene set, cannot tell them apart. Of the 1,140 singles with an essentiality entry, 907 also carry measurements; their stored fitness has median 0.679 where the measured mean has median 0.843, and the four bands in Figure 4a are the values of k . Those singles enter 141,099 closure doubles and 78,707 triples, and Section 3 measured what they cost. The 233 essentiality-only singles touch no closure double or triple, so they cost the identity nothing, but they teach a fitness model that 233 genes have a fitness of exactly 0 with no colony behind it.

H2: two fitness values in one record. A SynthLethDB single-gene record lists its gene twice, so its sorted gene set differs from the Costanzo single’s and the two are not merged; the trainer then finds two fitness values in one record,

Table 4: Records of the S3 pool touched by each hazard. Generated by `s3_closure_recompute.py`.

hazard	records
singles carrying an SGD essentiality entry (fitness 0 in the mean)	1,140
of which essentiality only, no measurement (stored fitness 0)	233
singles carrying a SynthLethDB entry (fitness 0)	3
records with two fitness values (masked by the trainer)	3
closure doubles touching an essentiality-tainted single	141,099
of which touching an essentiality-only single	0
triples touching an essentiality-tainted single	78,707
of which touching an essentiality-only single	0
closure doubles merged from several screens (p replaced by a t -test)	697,825
triples merged from several screens (p replaced by a t -test)	12,914
merged doubles significant in every source, not significant in the build	1,754
merged triples significant in every source, not significant in the build	406

0.0 and about 1.0. Three singles do this (YPL212C, YHL047C, YKL117W), and they killed the first S3 run at epoch 0 before the decode was changed to mask such rows. The 691 closure doubles with a SynthLethDB entry merge cleanly, a measured double-mutant fitness averaged with a 0, which is H1 again at order two.

H3: cross-screen singles in a within-screen identity. Eq. 1 is exact on a source row because f_a , f_b and f_{ab} were normalized together. The build’s single is a mean over every source that measured the gene, and its double is a mean over every screen of the pair. The single-screen stratum of Table 2 isolates this: a double whose fitness and score are one screen’s, against singles that are not, reaches $r = 0.339$.

H4: the merge replaces the p-value. Section 4 measured this. The count in Table 4 is 697,825 closure doubles and 12,914 triples whose stored p-value is a t -test with one to four degrees of freedom over duplicate scores, and the 1,754 doubles and 406 triples that every source called significant and the build does not.

H5: the source substituted 1.0 for a missing fitness. The released Kuzmin 2020 table has no query fitness on 99.6 percent of its digenic rows and the reported scores were computed with 1.0 in its place; Costanzo 2016 does the same on 2.1 percent of its rows. The build stores the score and stores whatever single-mutant fitness it has for the gene from other sources, so for these pairs no fitness in the build can reproduce the score, and a model asked to learn both is asked to learn an inconsistency the source created.

6 What this means for training on S3

The identity cannot be the mechanism. The S3 arm supervises fitness and interaction jointly on the hypothesis, stated as untested in the training plan, that a representation shaped by fitness helps the interaction head. If the mechanism were the model reading Eq. 2 off its fitness predictions, its ceiling on this build would be the $r = 0.23$ that the equation itself reaches on the stored values, below the $r = 0.40$ of the additive ridge that sees no fitness at all. Whatever the S3 seeds show at readout, they are not showing that; a gain would have to come from the representation, and a null would not refute the identity, which is exact in every source.

Per-order fitness is now logged, and per-order interaction was always logged. The trainer reports fitness for singles, doubles and triples and interaction for doubles and triples separately, so the readout can ask whether the model fits the essentiality-tainted singles, which hold a value no strain has, and whether the doubles’ fitness is learned better than their interaction, which this document predicts because the double fitness is a measurement and the double interaction is a cross-screen difference.

Every p-value in a merged record is unlabeled. Nothing in the training arm reads p-values, so no run is affected. Any analysis that does, a confidence-weighted loss, a filter to significant interactions, a precision-recall against called interactions, has to treat the 697,825 merged doubles and 12,914 merged triples as carrying no p-value, and can recover the sources’ p-values only by joining back to the raw tables by gene pair, as this document did.

Changes the build would need. Three, each at the step in Figure 1 that introduces the hazard. The essentiality conversion should not produce a fitness that the deduplicator can average with a measured one: either the converted 0 is kept as its own record with a perturbation type the merge key distinguishes, or the conversion is dropped from pools that also carry measured alleles of the gene. The deduplicator should keep the source p-values beside the merged score, or combine them by a rule that reads them (Fisher’s method on the source p-values, or a normal test on the merged score with the pooled source variance), rather than a t -test on two or three values. And a merged record should carry which screens went into it, so that a within-screen identity can be evaluated within a screen; the source tables carry the screen in the strain identifier, and the build drops it at the aggregation step.

7 Limits x

The recomputed p-values are not the sources' p-values. Section 4 is a ranking comparison. The normal test on the double's own colony SD orders the source p-values at $\rho = 0.984$ but calls 2.5 times as many interactions, because the source null pools an array-strain and a query-strain variance from control screens that were never released. No recomputation from the build or from the released tables can recover those; the statement is that the source p is reproducible in rank, not in value.

The measured-mean restoration assumes exactly one 0 in the mean. The $k/(k - 1)$ correction in Section 5 assumes a tainted single merged exactly one essentiality entry of value 0 with $k - 1$ measured entries. The `num_duplicates` field gives k and the source list confirms one essentiality entry for every tainted single, but the individual measured values are gone after the merge, so the restored mean is exact only if the converter wrote exactly 0. It did (`CompositeFitnessConverter`), and the two checked cases (YER048W-A, stored 0.452 from 0.904 and 0; YDR292C, stored 0.630 over $k = 7$) agree.

The within-screen control covers the digenic identity only. The raw Kuzmin tables carry the double-mutant query fitness and the triple fitness on a trigenic row, but the two digenic control-screen terms of Eq. 2 sit on other rows of the same table and are matched by strain identifier, which this document did not do. The trigenic identity is therefore checked in the build only, and its $r = 0.230$ has no within-screen counterpart here; the Kuzmin supplementary text reports replicate correlations of 0.74 to 0.81 for the adjusted trigenic score (`si/si1.md` line 171), which bounds what a within-screen check could reach.

Four colonies is a documented representative, not a per-record count. The null standard deviation $\sigma_{ab}/\sqrt{4}$ uses the colony count the loaders record (Costanzo 2016: "4 replicate colonies per double mutant"; the Kuzmin loader back-solved $n = 4$ against the reported p-values). Only 41,370 of the closure doubles carry an explicit `n_samples`; the rest use the same value. A different n rescales every recomputed p monotonically and changes no ranking in Table 3.

Nothing here reads a training run. Every number is a property of the pool. The three S3 seeds (IGB `mmli` 2409033, 2409031, 2409032) were running or queued when this was written and are read out elsewhere.